Chapter 38

38 Carnot, the Limit of Efficiency for Heat Engines 2.4.4, 2.4.5

Last lecture we studied a somewhat realistic heat engine. We found we got something like a 56.5% efficiency. But is this really realistic? Could you build such an engine? We need a way to tell what the maximum efficiency for a heat engine would be. Then, if our design exceeds this maximum efficiency, we know we have a mistake in our design. Our goal in a heat engine problem is always to find for each process the values of ΔE_{int}, Q, W, and for the whole engine W_{eng}, Q_h, Q_c. We also want to find the ideal gas state variables P, V, n, and T at the transition points between processes. But we need the maximum efficiency to know if our result is realistic. Let's use all of these techniques to find our maximum efficiency.

Fundamental Concepts

- The Carnot cycle represents the maximum theoretical limit of efficiency for a heat engine.
- The efficiency of the Carnot cycle depends only on the two extreme temperatures $\eta_C = 1 - \frac{T_c}{T_h}$.
- An engine design that is more efficient that a Carnot engine must have a fundamental flaw in it's calculations. Question 123.19.1

38.1 Carnot's Theorem

We now know that we can design actual engines that power cars and useful machines by combining several of our special thermodynamic process into larger, more complex processes that form a cycle. You might guess that there are more than just isochoric, isobaric, isothermal, and adiabatic processes that exist. A good question is what processes should we combine to form our engine? Past researchers have asked this question over and over and many new engines have been developed. Of course we want our engines to be as efficient as possible, and the second law of thermodynamics says the engines can't be 100% efficient. But is there a "most efficient" engine, one that no other engine can beat in efficiency? If there is, we could use this to judge new engine designs. Any engine with a better efficiency would be a mistake that won't work, and that approach this maximum efficiency would be a possible improvement in technology. There is such a maximally efficient engine.

Carnot was a French Engineer. He thought of an engine that would operate on an ideal reversible cycle. He hit upon the most efficient cycle possible. Sadly, it is not possible to create his engine in practice. Carnot's engine assumed no friction and ideal thermodynamic processes. So why is it useful to know about Carnot's engine design? The Carnot cycle provides us with a useful upper limit for all practical engines. To tell if a design is practical, we can compare it to the Carnot cycle. If our design looks like it will outperform the Carnot engine, we know we have done something wrong. Or conversely, if our system design requires an engine to be better than a Carnot engine, we need to go back to the drawing board.

Knowing just this much we can state Carnot's theorem.

No real heat engine operating between two energy reservoirs can be more efficient than a Carnot engine operating between the same two reservoirs

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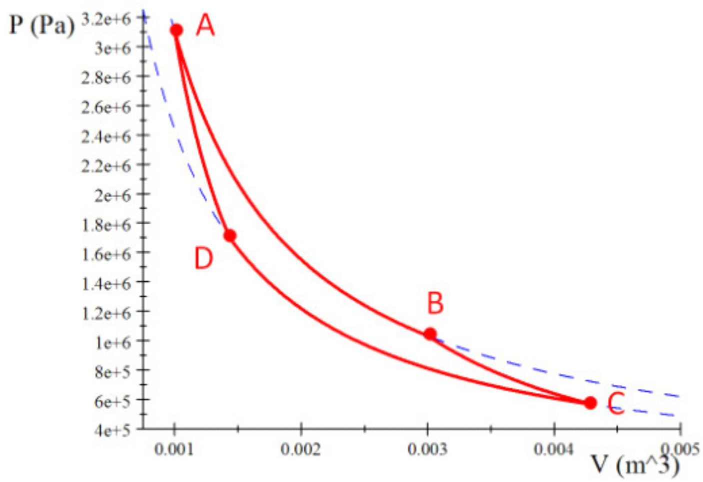
All real engines are less efficient than a Carnot engine because they do not operate through a reversible cycle. The efficiency of a real engine is reduced by friction, energy losses through conduction, etc.

38.2 The Carnot Cycle

Carnot found a set of special processes that when combined made a theoretical heat engine. And he proved mathematically that this particular combination is theoretically the most efficient possible. Let's take a look at this engine design.

The Carnot cycle is shown in the figure. It consists of the following parts:

1. Process $A \rightarrow B$ is an isothermal expansion
2. Process $B \rightarrow C$ is an adiabatic expansion
3. Process $C \rightarrow D$ is an isothermal contraction
4. Process $D \rightarrow A$ is an adiabatic compression



Let's take an example. Let's choose a Carnot engine that operates between the same two extreme temperatures as our Otto cycle example.

$$
\begin{aligned}
T_c &= 273\,\mathrm{K} \\
T_h &= 1023\,\mathrm{K}
\end{aligned}
$$

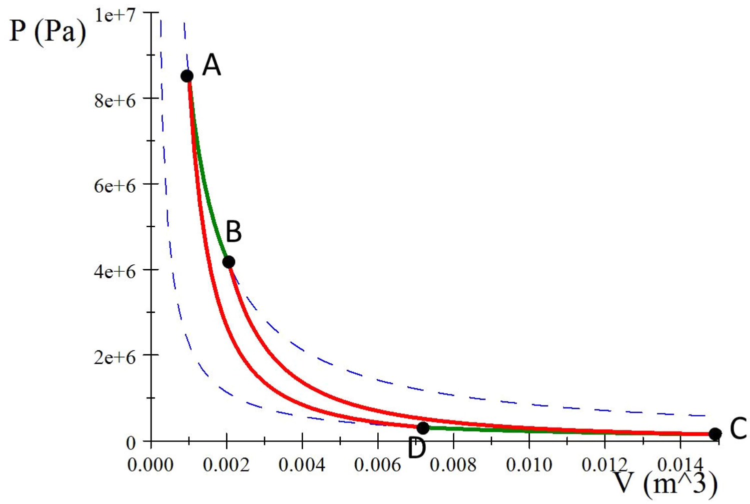
Let's suppose the carnot engine is made from a piston with a monotonic idea gas. Let's choose the minimum and maximum volumes to be

$$
\begin{aligned}
\cancel{V}_A &= 0.001\,\mathrm{m}^3 \\
\cancel{V}_C &= 0.015\,\mathrm{m}^3
\end{aligned}
$$

and let's have

$$n = 1\,\mathrm{mol}$$

of gas. For these numbers, our graph looks like this:

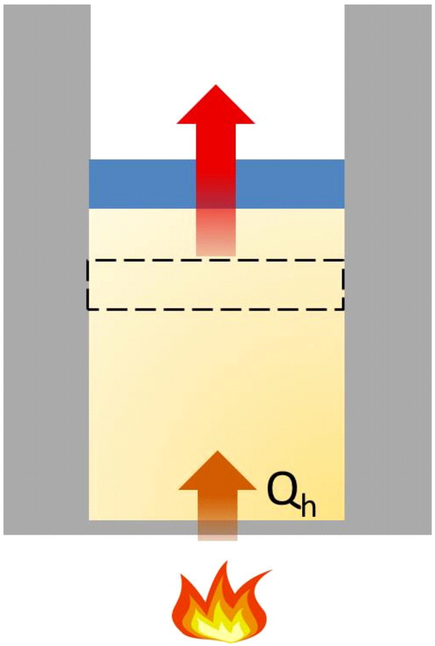


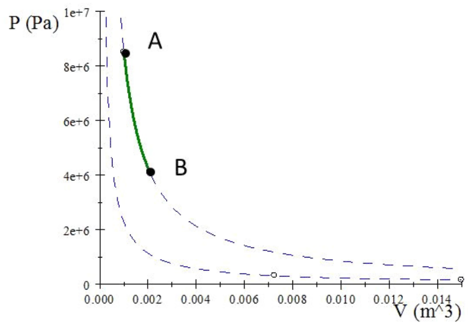
Our job is to find P, $\forall$, T, W, Q, and ΔE_{int} for each part of the cycle and Q_h, Q_c, and W_{eng} for the whole cycle.

Let's take each process separately like we did for the Otto cycle.

38.3 Process $A \rightarrow B$ isothermal expansion

The gas is placed in contact with the high temperature reservoir, T_h. The gas absorbs heat $|Q_h|$. The gas does work W_{AB} in raising the piston





Remember for isothermal processes we can write

$$\frac{P_i \mathcal{V}_i}{P_f \mathcal{V}_f} = \frac{nRT_i}{nRT_i}$$

because the temperature does not change, so

$$P_i \mathcal{V}_i = P_f \mathcal{V}_f$$

Along Path AB this would be

$$P_A \mathcal{V}_A = P_B \mathcal{V}_B$$

and we know for an isothermal process that

$$W_{AB} = nRT_A \ln\left(\frac{\mathcal{V}_A}{\mathcal{V}_B}\right)$$

and that

$$\Delta E_{int} = 0$$

so

$$|Q_{AB}| = |-W_{AB}|$$

and we can identify this as $|Q_h|$ and $T_A = T_h$

We can complete state A using the ideal gas law

$$\begin{aligned} P_A &= \frac{nRT_h}{\mathcal{V}_A} = \frac{(1\,\mathrm{mol})\left(8.314\frac{\mathrm{J}}{\mathrm{mol\,K}}\right)(1023\,\mathrm{K})}{0.001\,\mathrm{m}^3} \\ &= 8.505\,2\times10^6\,\mathrm{Pa} \end{aligned}$$

To complete state B we need the volume and the pressure. At point B both the adiabat and the isotherm have the same pressure and volume. So we know

$$P_B = \frac{nRT_h}{\mathcal{V}_B}$$

and

$$P_B = \frac{K_{BC}}{\mathcal{V}_B^{\gamma}}$$

where the constant $K_{BC} = P_B\mathcal{V}_B^{\gamma}$ from our adiabatic equations. But we don't know K_{BC}, P_B, or $\mathcal{V}_B$

We do know that at C

$$P_C = \frac{nRT_c}{\mathcal{V}_C}$$

and that

$$P_C = \frac{K_{BC}}{\mathcal{V}_c^{\gamma}}$$

and at point C and we do know $\mathcal{V}_C$, so we can find K_{BC}. Set the two expressions for P_C equal

$$\frac{nRT_c}{\mathcal{V}_C} = \frac{K_{BC}}{\mathcal{V}_C^{\gamma}}$$

$$\begin{aligned} K_{BC} &= \frac{nRT_c}{\mathcal{V}_C}\mathcal{V}_C^{\gamma} \\ &= nRT_c\mathcal{V}_C^{\gamma-1} \end{aligned}$$

so

$$\begin{aligned} K_{BC} &= nRT_c\mathcal{V}_C^{\gamma-1} = (1\,\mathrm{mol})\left(8.314\frac{\mathrm{J}}{\mathrm{mol\,K}}\right)(273\,\mathrm{K})\left(0.015\,\mathrm{m}^3\right)^{\frac{2}{3}} \\ &= 138.\,05\,\mathrm{J\,m}^2 \end{aligned}$$

Then, moving back to point B we can set the two equations for P_B equal

$$\begin{aligned} P_B &= \frac{nRT_h}{\mathcal{V}_B} \\ P_B &= \frac{K_{BC}}{\mathcal{V}_B^{\gamma}} \end{aligned}$$

to get

$$\frac{nRT_h}{\mathcal{V}_B} = \frac{K_{BC}}{\mathcal{V}_B^{\gamma}}$$

then

$$\frac{V_B^{\gamma}}{V_B} = \frac{K_{BC}}{nRT_h}$$
$$V_B^{\gamma-1} = \frac{K_{BC}}{nRT_h}$$

which gives us an awkward

$$V_B = \sqrt[\gamma-1]{\frac{K_{BC}}{nRT_h}}$$

but it works.

$$\begin{aligned} V_B &= \sqrt[\frac{2}{3}]{\frac{138.05\,\mathrm{J\,m^2}}{(1\,\mathrm{mol})\left(8.314\frac{\mathrm{J}}{\mathrm{mol\,K}}\right)(1023\,\mathrm{K})}} \\ &= 2.0679 \times 10^{-3}\,\mathrm{m^3} \end{aligned}$$

And we are back to the ideal gas law to find the pressure

$$\begin{aligned} P_B &= \frac{nRT_B}{V_B} \\ &= \frac{(1\,\mathrm{mol})\left(8.314\frac{\mathrm{J}}{\mathrm{mol\,K}}\right)(1023\,\mathrm{K})}{2.0679 \times 10^{-3}\,\mathrm{m^3}} \\ &= 4.1130 \times 10^{6}\,\mathrm{Pa} \end{aligned}$$

Armed with all of this, we can compute the work and heat.

$$W_{AB} = nRT_h \ln\left(\frac{V_A}{V_B}\right)$$

$$\begin{aligned} W_{AB} &= (1\,\mathrm{mol})\left(8.314\frac{\mathrm{J}}{\mathrm{mol\,K}}\right)(1023\,\mathrm{K})\ln\left(\frac{0.001\,\mathrm{m^3}}{2.0679 \times 10^{-3}\,\mathrm{m^3}}\right) \\ &= -6179.3\,\mathrm{J} \end{aligned}$$

We can find

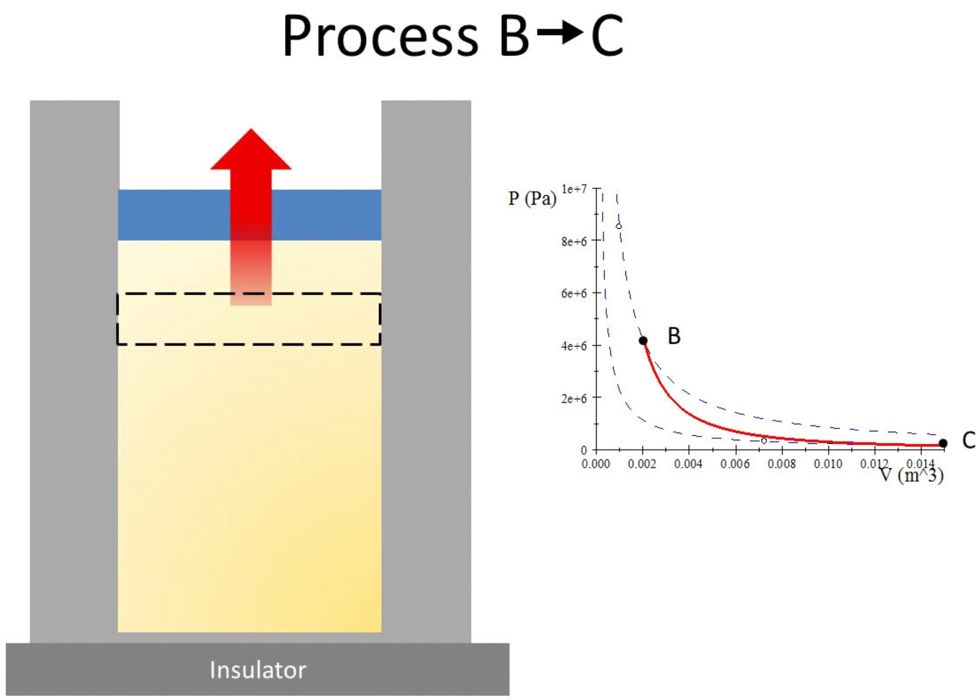
State	**T** (K)	**P** (Pa)	V (m^3)	**n** (mol)	–	–
A	1023	8.5052×10^{6}	0.001	1	–	–
B	1023	4.1130×10^{6}	2.0679×10^{-3}	1	–	–
Process	**Q** (J)	$\mathbf{W}_{int}$ (J)	$\mathbf{\Delta E}_{int}$ (J)	$\mathbf{W}_{eng}$ (J)	$\mathbf{Q}_h$	$\mathbf{Q}_c$
AB	6179.3	−6179.3	0	6179.3	6179.3	0

38.4 Process $B \rightarrow C$ adiabatic expansion

The base of the cylinder is replaced by a thermally nonconducting wall. No energy enters or leaves the system by heat.

$$Q = 0$$

The temperature falls from T_h to T_c. The gas does work W_{BC}.



Remember that for an adiabatic process

$$P_A \cancel{V}_A^{\gamma} = P_B \cancel{V}_B^{\gamma}$$

and we usually find a more convenient method for calculating ΔE_{int}. We choose an isochoric path, even though it isn't right. But ΔE_{int} is the same for any path so

$$\Delta E_{int} = nC_V \Delta T$$

We can find the pressure at C using the ideal gas law

$$\begin{aligned} P_C &= \frac{nRT_c}{\cancel{V}_C} = \frac{(1\,\mathrm{mol})\left(8.314\frac{\mathrm{J}}{\mathrm{mol\,K}}\right)(273\,\mathrm{K})}{0.015\,\mathrm{m}^3} \\ &= 1.5131 \times 10^5\,\mathrm{Pa} \end{aligned}$$

and

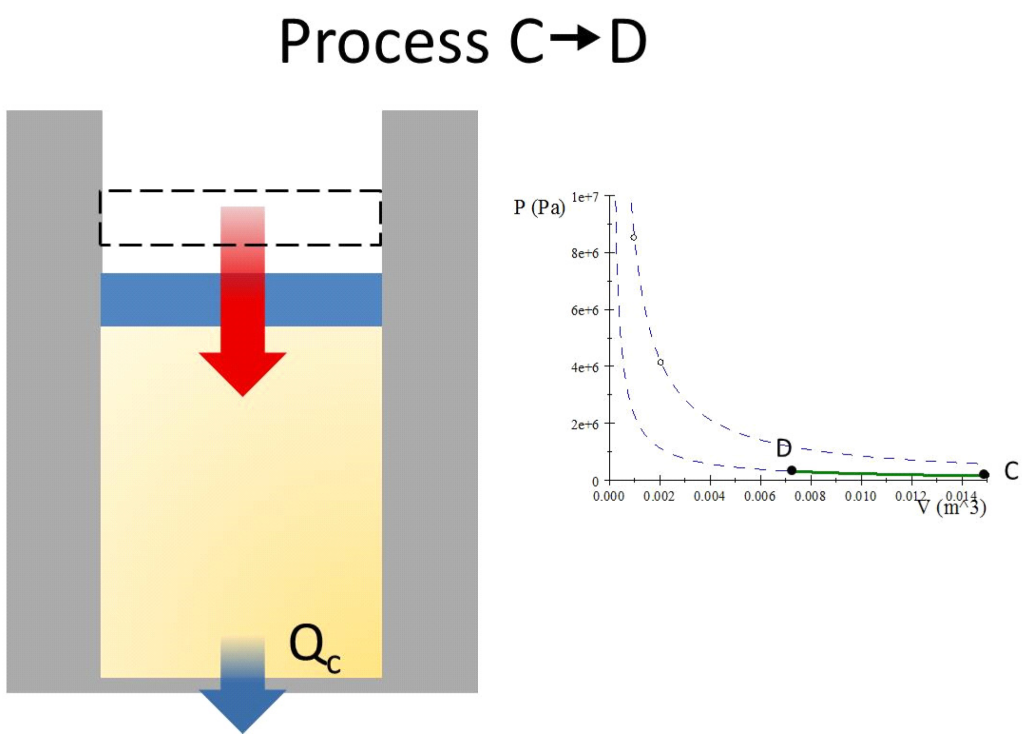
$$\begin{aligned} \Delta E_{int} &= (1\,\mathrm{mol})\frac{3}{2}\left(8.314\frac{\mathrm{J}}{\mathrm{mol\,K}}\right)(273\,\mathrm{K} - 1023\,\mathrm{K}) \\ &= -9353.3\,\mathrm{J} \end{aligned}$$

we can summarize all this.

State	**T** (K)	**P** (Pa)	$\not{V}$ (m^3)	**n** (mol)	–	–
B	1023	4.1130×10^6	2.0679×10^{-3}	1	–	–
C	273	1.5131×10^5	$0.015\,\mathrm{m}^3$	1	–	–
Process	**Q** (J)	$\mathbf{W}_{int}$ (J)	$\mathbf{\Delta E}_{int}$ (J)	$\mathbf{W}_{eng}$ (J)	$\mathbf{Q}_h$	$\mathbf{Q}_c$
BC	0	-9353.3	-9353.3	9353.3	0	0

38.5 Process $C \to D$ isothermal compression

The gas is placed in contact with the cold temperature reservoir. The gas expels energy $|Q_c|$ and work W_{CD} is done on the gas.



The process is isothermal, so $T_C = T_D$ and $nRT_C = nRT_D$ so we have

$$P_C \not{V}_C = P_D \not{V}_D \tag{38.1}$$

and

$$W_{CD} = nRT_c \ln\left(\frac{\not{V}_C}{\not{V}_D}\right) \tag{38.2}$$

again

$$\Delta E_{int} = 0 \tag{38.3}$$

so

$$|Q_{CD}| = |-W_{CD}| \tag{38.4}$$

and we can identify this as $|Q_c|$

We again need to find a volume, $\not{V}_D$ and we can use the same method as before, finding $\not{V}_B$. At position D

$$P_D = \frac{nRT_c}{\not{V}_D}$$

and

$$P_D = \frac{K_{DA}}{\not{V}_D^{\gamma}}$$

but we don't know K_{DA}, P_D, or $\not{V}_D$

But we also know that at A

$$\begin{aligned} P_A &= \frac{nRT_h}{\not{V}_A} \\ P_A &= \frac{K_{DA}}{\not{V}_A^{\gamma}} \end{aligned}$$

and at point A and we do know $\not{V}_{A,}$ so we can find K_{DA}

$$\frac{nRT_h}{\not{V}_A} = \frac{K_{DA}}{\not{V}_A^{\gamma}}$$

so

$$K_{DA} = nRT_h \not{V}_A^{\gamma-1}$$

or

$$\begin{aligned} K_{DA} &= (1\,\mathrm{mol}) \left(8.314 \frac{\mathrm{J}}{\mathrm{mol\,K}}\right) (1023\,\mathrm{K}) \left(0.001\,\mathrm{m}^3\right)^{\frac{2}{3}} \\ &= 85.052\,\mathrm{J\,m}^2 \end{aligned}$$

Then moving back to point D we have

$$\frac{nRT_c}{\not{V}_D} = \frac{K_{DA}}{\not{V}_D^{\gamma}}$$

and

$$\not{V}_D = \sqrt[\gamma-1]{\frac{K_{DA}}{nRT_c}}$$

Numerically we get

$$\begin{aligned} \not{V}_D &= \sqrt[\frac{2}{3}]{\frac{85.052\,\mathrm{J\,m}^2}{(1\,\mathrm{mol}) \left(8.314 \frac{\mathrm{J}}{\mathrm{mol\,K}}\right) (273\,\mathrm{K})}} \\ &= 7.2537 \times 10^{-3}\,\mathrm{m}^3 \end{aligned}$$

And we are back to the ideal gas law to find the pressure

$$\begin{aligned} P_D &= \frac{(1\,\text{mol})\left(8.314\frac{\text{J}}{\text{mol K}}\right)(273\,\text{K})}{7.2537\times10^{-3}\,\text{m}^3} \\ &= 3.1291\times10^5\,\text{Pa} \end{aligned}$$

And so we can find the work as

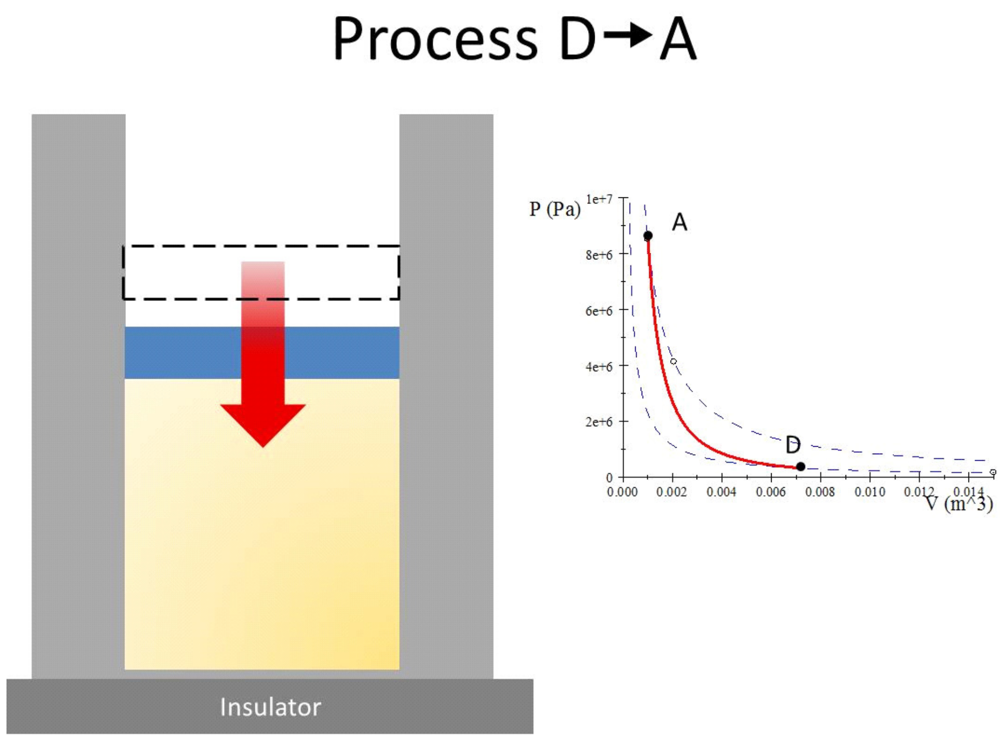
$$\begin{aligned} W_{CD} &= (1\,\text{mol})\left(8.314\frac{\text{J}}{\text{mol K}}\right)(273\,\text{K})\ln\left(\frac{0.015\,\text{m}^3}{7.2537\times10^{-3}\,\text{m}^3}\right) \\ &= 1649.0\,\text{J} \end{aligned}$$

And our summary is as follows

State	**T** (K)	**P** (Pa)	$\mathcal{V}$ (m^3)	**n** (mol)	–	–
C	273	1.5131×10^5	$0.015\,\text{m}^3$	1	–	–
D	273	3.1291×10^5	7.2537×10^{-3}	1	–	–
Process	**Q** (J)	$\mathbf{W}_{int}$ (J)	$\mathbf{\Delta E}_{int}$ (J)	$\mathbf{W}_{eng}$ (J)	$\mathbf{Q}_h$	$\mathbf{Q}_c$
CD	−1649.0	1649.0	0	0	0	1649.0

38.6 Process $D \rightarrow A$ adiabatic compression

The gas cylinder is again placed against a thermally nonconducting wall so no heat is exchanged with the surroundings. The temperature of the gas increases from T_c to T_h.The work done on the gas is W_{DA}



We know $Q = 0$ and that $\Delta E_{int} = W$. Again we can use

$$\Delta E_{int} = nC_V \Delta T$$

$$\begin{aligned}\Delta E_{int} &= (1\,\mathrm{mol})\,\frac{3}{2}\left(8.314\frac{\mathrm{J}}{\mathrm{mol\,K}}\right)(1023\,\mathrm{K} - 273\,\mathrm{K}) \\ &= 9353.3\,\mathrm{J}\end{aligned}$$

so then

State	**T** (K)	**P** (Pa)	$\mathbf{V}$ $\left(\mathrm{m}^3\right)$	**n** (mol)	–	–
D	273	3.1291×10^5	7.2537×10^{-3}	1	–	–
A	1023	3.1×10^6	0.001	1	–	–
Process	**Q** (J)	$\mathbf{W}_{int}$ (J)	$\mathbf{\Delta E}_{int}$ (J)	$\mathbf{W}_{eng}$ (J)	$\mathbf{Q}_h$	$\mathbf{Q}_c$
DA	0	9353.3	9353.3	−9353.3	0	0

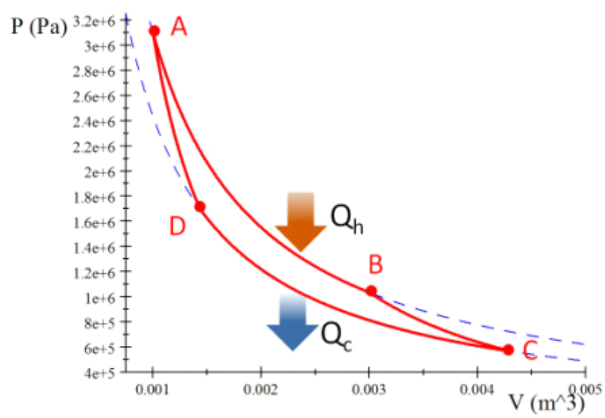
All together we have

State	**T** (K)	**P** (Pa)	$\mathbf{V}$ $\left(\mathrm{m}^3\right)$	**n** (mol)
A	1023	8.5052×10^6	0.001	1
B	1023	4.1130×10^6	2.0679×10^{-3}	1
C	273	1.5131×10^5	0.015	1
D	273	3.1291×10^5	7.2537×10^{-3}	1

And our internal energy, heat, and work are

Process	**Q** (J)	$\mathbf{W}_{int}$ (J)	$\mathbf{\Delta E}_{int}$ (J)	$\mathbf{W}_{eng}$ (J)	$\mathbf{Q}_h$	$\mathbf{Q}_c$
AB	6179.3	−6179.3	0	6179.3	6179.3	0
BC	0	−9353.3	−9353.3	9353.3	0	0
CD	−1649.0	1649.0	0	−1649.0	0	1649.0
DA	0	9353.3	9353.3	−9353.3	0	0
Total	4530.3	−4530.3	0	4530.3	6179.3	1649.0

We can see that process $A \to B$ provides Q_h and process $C \to D$ provides Q_C.Let's label this in our PV diagram.



We can also find the efficiency of the Carnot cycle. The net work is W_{ABCDA} and the efficiency is

$$\eta = \frac{W_{eng}}{|Q_h|} = 1 - \frac{|Q_c|}{|Q_h|}$$

No heat is transferred during the adiabatic processes, so the only heat transfer is only during the isothermal processes, we have

$$\begin{aligned} \eta &= 1 - \frac{\left| nRT_c \ln\left(\frac{V_C}{V_D}\right)\right|}{\left| nRT_h \ln\left(\frac{V_A}{V_B}\right)\right|} \\ &= 1 - \frac{\left| T_c \ln\left(\frac{V_C}{V_D}\right)\right|}{\left| T_h \ln\left(\frac{V_A}{V_B}\right)\right|} \end{aligned}$$

Using the adiabatic temperature-volume relationships we can find expressions for the hot and cold temperatures

$$T_h V_A^{\gamma-1} = T_c V_D^{\gamma-1}$$

and

$$T_h \mathcal{V}_B^{\gamma-1} = T_c \mathcal{V}_C^{\gamma-1}$$

We could divide these two equations to get

$$\frac{T_h \mathcal{V}_A^{\gamma-1}}{T_h \mathcal{V}_B^{\gamma-1}} = \frac{T_c \mathcal{V}_D^{\gamma-1}}{T_c \mathcal{V}_C^{\gamma-1}}$$

or

$$\frac{\mathcal{V}_A^{\gamma-1}}{\mathcal{V}_B^{\gamma-1}} = \frac{\mathcal{V}_C^{\gamma-1}}{\mathcal{V}_D^{\gamma-1}}$$

which gives us

$$\sqrt[\gamma-1]{\frac{\mathcal{V}_A^{\gamma-1}}{\mathcal{V}_B^{\gamma-1}}} = \sqrt[\gamma-1]{\frac{\mathcal{V}_C^{\gamma-1}}{\mathcal{V}_D^{\gamma-1}}}$$

But if we take both sides to the $\gamma - 1$ power we have just

$$\frac{\mathcal{V}_A}{\mathcal{V}_B} = \frac{\mathcal{V}_D}{\mathcal{V}_C}$$

Using this we find that our efficiency equation is

$$\begin{aligned} \eta &= 1 - \frac{\left| T_c \ln\left(\frac{\mathcal{V}_C}{\mathcal{V}_D}\right)\right|}{\left| T_h \ln\left(\frac{\mathcal{V}_D}{\mathcal{V}_C}\right)\right|} \\ &= 1 - \frac{\left| T_c \ln\left(\frac{\mathcal{V}_C}{\mathcal{V}_D}\right)\right|}{\left| -T_h \ln\left(\frac{\mathcal{V}_C}{\mathcal{V}_D}\right)\right|} \end{aligned}$$

or, removing the minus sign because of the absolute values

$$\eta = 1 - \frac{T_c}{T_h}$$

where the temperatures, of course, are in Kelvin. This means that for this engine

$$\frac{|Q_c|}{|Q_h|} = \frac{T_c}{T_h}$$

and

$$\eta_C = 1 - \frac{T_c}{T_h} \tag{38.5}$$

This is the general form for the efficiency of a Carnot cycle. Then for our example

$$\begin{aligned} \eta_C &= 1 - \frac{273\,\mathrm{K}}{1023\,\mathrm{K}} \\ &= 0.733\,14 \end{aligned}$$

This is a great review of how to use our thermodynamic equations for special process, even though we can't build a Carnot engine. But more importantly, we can judge the efficacy of engine designs. For example, let's take our Otto cycle from last time. Our design gave

$$\begin{aligned}\eta_{otto} &= 1-\frac{1}{\left(\frac{0.0005}{6.25\times10^{-5}}\right)^{1.4-1}}\\ &= 0.56472\end{aligned}$$

So the Otto cycle example might be achievable. Of course, we can tell that our Otto cycle is still an idealization. We recognize that we neglected all friction, and assumed perfect adiabatic processes. But we can see that there is some hope of making our Otto cycle-based engines more efficient. But no Otto cycle that operates between the same $T_c = 273\,\mathrm{K}$ and $T_h = 1023\,\mathrm{K}$ can ever be better than 73% effective because that is the maximum efficiency for a heat engine operating between these two temperatures.

This achieved our goal. We have a way to find the maximum efficiency for a heat engine. And along the way we made several statements of the second law of thermodynamics. But if you were left with the feeling that we really didn't determine what the second law meant by looking at the physics of what is happening to our gas, you are right. Let's take that on in our last lecture.